Transformers in Neural Networks

The Architecture Behind ChatGPT

African Institute for Mathematical Sciences (AIMS)

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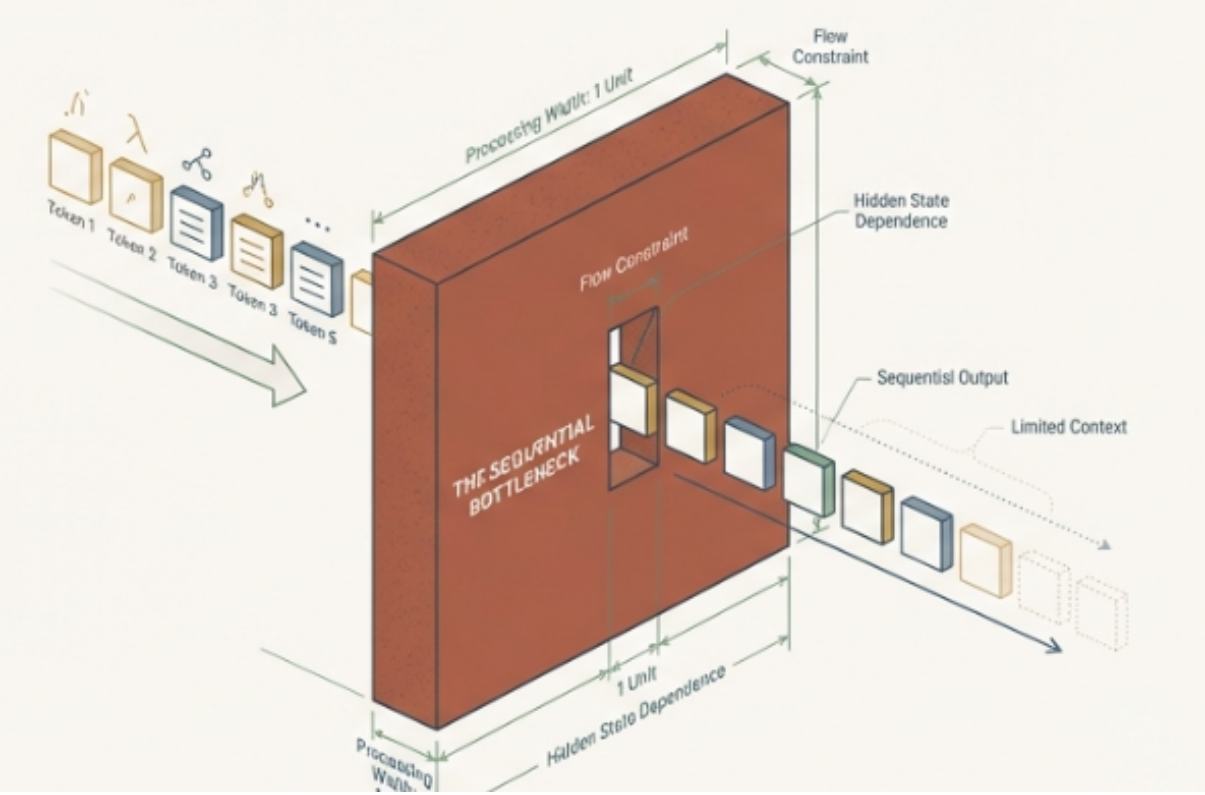
Group 1

Group Members: Adam, Ayanda, Blanche, Blessing, Eunice, Laura

Presentation Outline

- **The Motivation**: Limitations of RNNs and LSTMs
- **The Core Innovation**: The Attention Mechanism (Q, K, V)
- **Inside the Model**: Anatomy and Multi-Head Attention
- **Practical Demo**: Sentiment Analysis on IMDB Reviews
- **Performance**: Transformer vs. RNN Evaluation
- **Real-World Impact**: Modern Applications (ChatGPT, BERT, etc.)
- **Conclusion**

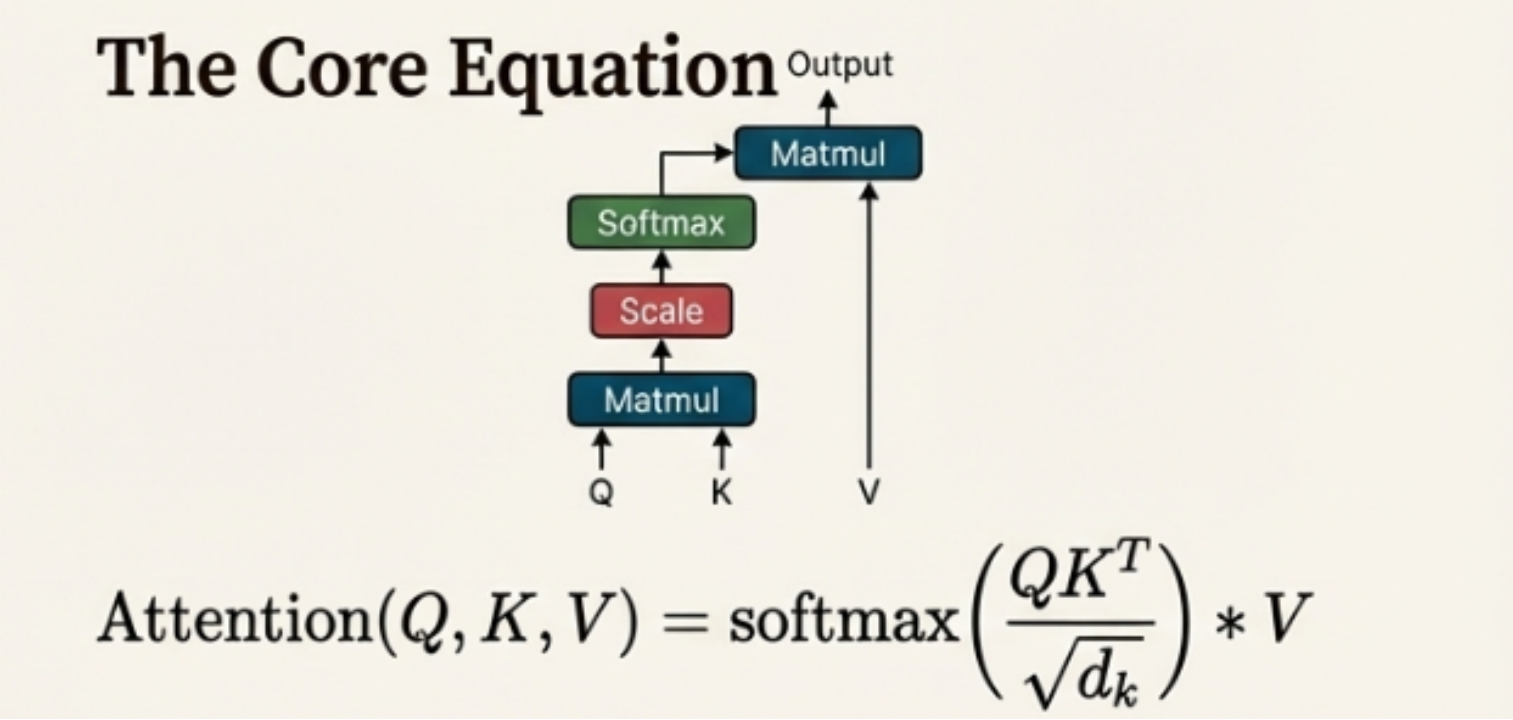
Before Transformers: The Limitations



Old Approach: RNN/LSTM

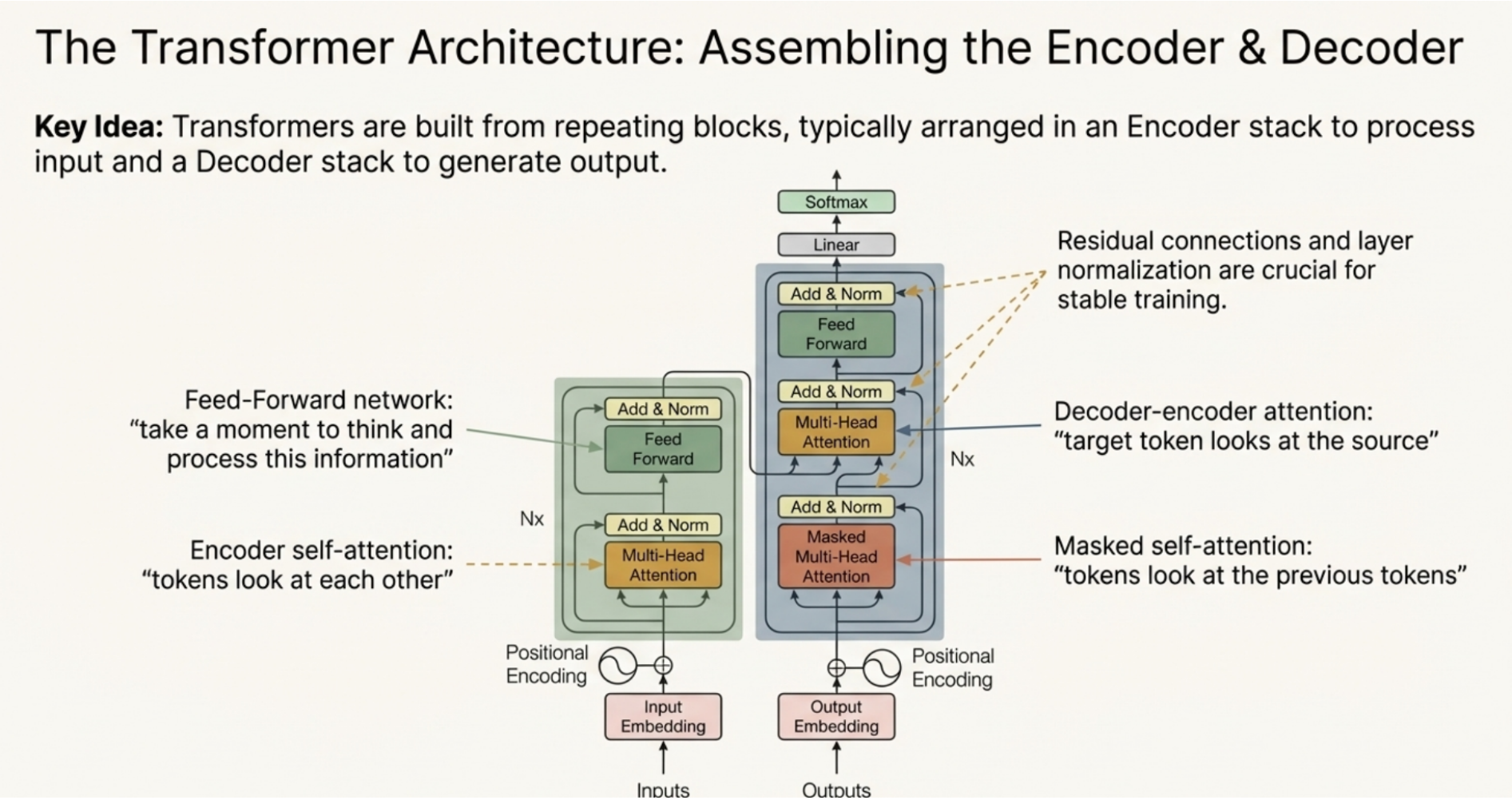
- × **Sequential Processing** (Slow)
- × **Memory Loss** (Short context)
- × **No Parallelization**

The Big Innovation: Attention Mechanism



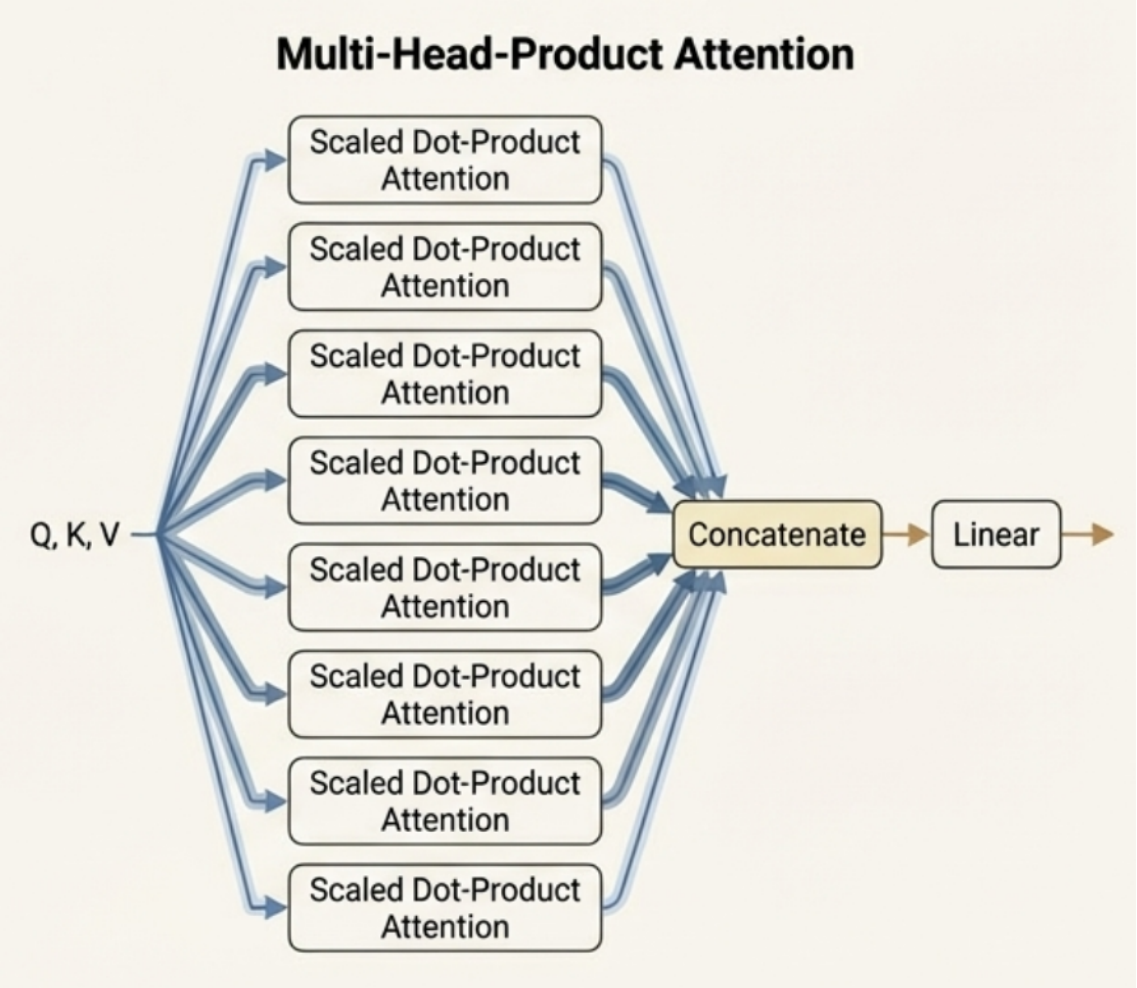
- **Query (Q):** What you're searching for
- **Key (K):** Keywords in documents
- **Value (V):** Actual content

Anatomy of a Transformer



The Architecture of the Transformer

Many Perspectives are Better Than One: Multi-Head Attention



Multi-Head Attention Architecture

Why Multiple Heads?

- Each head learns **different patterns**.
- Like having multiple expert readers analyzing the same text.

The Result

- Captures **longer dependencies**.
- Improves model stability and performance.

Does It Work? Sentiment Analysis Demo

Task: Movie Review Classification

Dataset: IMDB Reviews (Positive/Negative)

Our Transformer:

- Accuracy: **84.3%**
- Training: Fast (parallel)
- Beats RNN (50.1%)

Example Predictions:

✓ "Fantastic movie!"
→ Positive (95%)

✓ "Terrible, don't watch"
→ Negative (89%)

Backup: Technical Details I

Positional Encoding Formula:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

Backup: Technical Details II

Our Model Configuration:

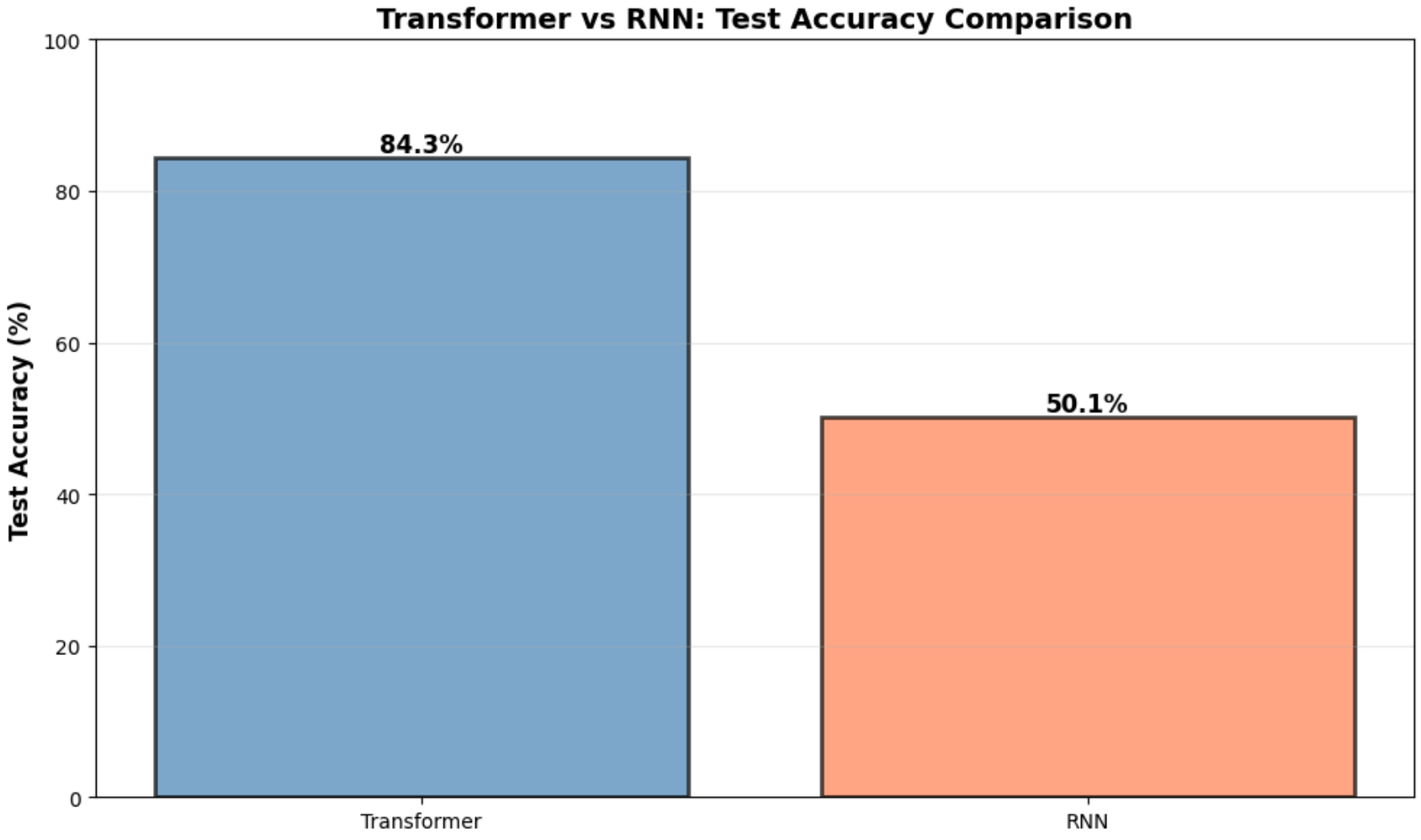
- Embedding dimension: 64
- Number of heads: 4
- Feed-forward dimension: 128
- Vocabulary size: 10,000
- Max sequence length: 200
- Dropout rate: 0.1

Backup: Technical Details III

Training Details:

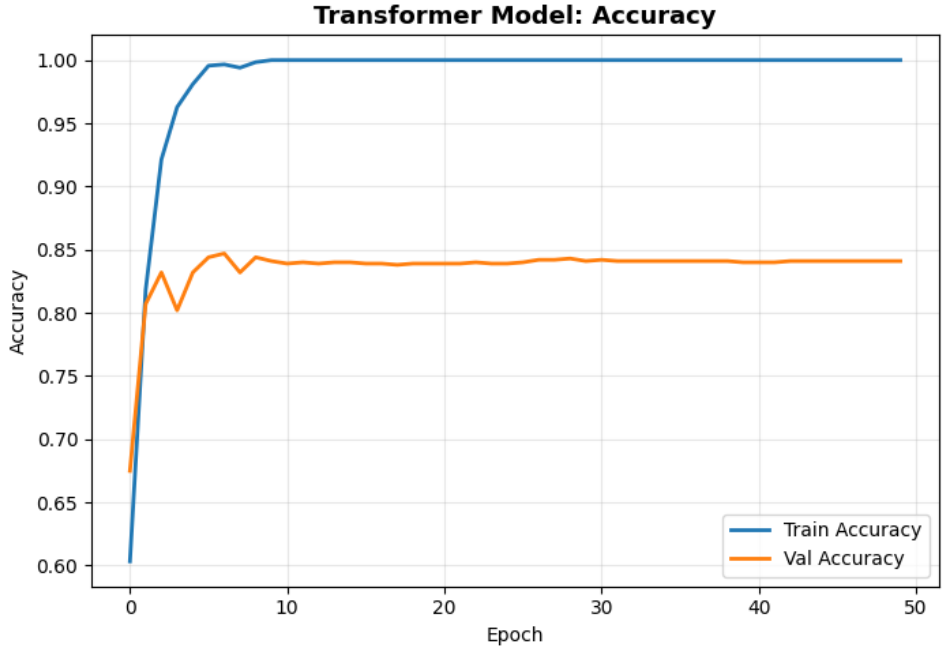
- Dataset: IMDB Movie Reviews
- Training samples: 25,000
- Test samples: 25,000
- Batch size: 64
- Optimizer: Adam
- Epochs: 50

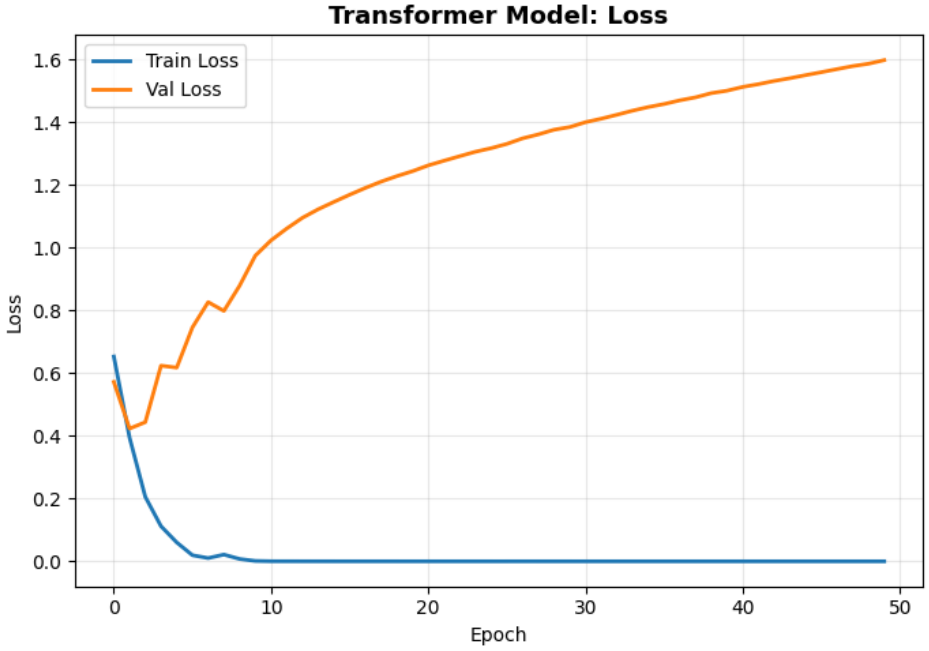
Transformer's Evaluation



Transformer vs RNN

Transformer's Evaluation



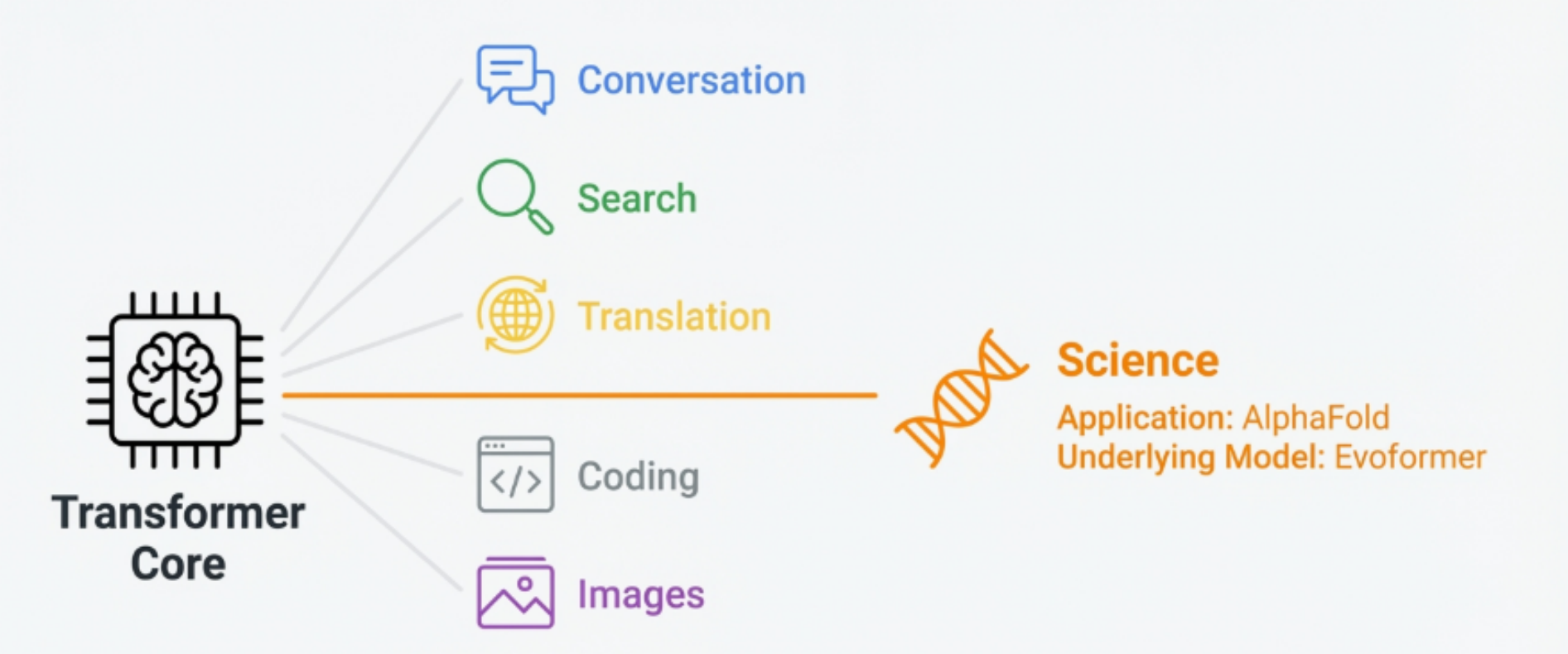


Transformer's Model: Accuracy and Loss

Transformers Power Everything

Domain	Application	Model
Conversation	ChatGPT, Claude	GPT-4
Search	Google Search	BERT
Translation	Google Translate	Transformer
Coding	GitHub Copilot	Codex
Images	DALL-E	Vision Transformer
Science	AlphaFold	Evoformer

Transformers Power Everything



The Applications of the Transformer

Why Transformers Dominate

Parallelization

- Process all words simultaneously
- 100× faster than RNNs
- Efficient on GPUs

Long-Range

- Direct word connections
- No information loss
- Handles 1000s of tokens

Scalability

- Bigger = Better
- Transfer learning
- State-of-the-art

The Winning Formula:

Attention + Parallelization + Scale =
Revolutionary AI

Conclusion

Summary

Solved sequential processing through **attention mechanism** and **parallel processing**

Future Directions:

- More efficient transformers
- Longer context windows
- Multimodal AI

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Questions?

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Thank you for your attention!